4.5.1 Photons

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the particulate nature (photon model) of electromagnetic radiation	
(b)	photon as a quantum of energy of electromagnetic radiation	
(c)	energy of a photon; $E = hf$ and $E = \frac{hc}{\lambda}$	
(d)	the electronvolt (eV) as a unit of energy	
(e)	**(i)** using LEDs and the equation $eV = \frac{hc}{\lambda}$ to estimate the value of Planck constant h	No knowledge of semiconductor theory is required. HSW11
	(ii) Determine the Planck constant using different coloured LEDs.	**PAG6**